

Curriculum Vitae



Dr. Alkesh Agrawal

<https://www.scopus.com/authid/detail.uri?authorId=57194016744>

<https://orcid.org/0000-0001-6326-7560?lang=en>

<https://scholar.google.co.in/citations?user=Yc6hSfcAAAAJ&hl=en>

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Mob. : 9935363326

DoB: 26th March 1976

Marital Status: Married

Residential Address: 21/1, Amrapali Vihar, Rajni Khand Road, Telibagh, Lucknow-25

Professional Experience

Shri Ramswaroop Memorial University, Institute of Technology, Lucknow-Deva Road, Barabanki, Uttar Pradesh.

11-07-2013 to Present | **Associate Professor, HoD**

(Electronics and Communication Engineering)

President: IIC-SRMU (Institution's Innovation Council) U-0753

Shri Ramswaroop Memorial University, Institute of Technology, Lucknow-Deva Road, Barabanki, Uttar Pradesh.

01-07-2012 to 10-07-2013 | **Assistant Professor**

(Electronics and Communication Engineering)

Shri Ramswaroop Memorial College of Engineering and Management, Lucknow, Uttar Pradesh.

23-07-2009 to 30-06-2012 | **Assistant Professor**

Head of the Department

(Electronics and Instrumentation)

Shri Ramswaroop Memorial College of Engineering and Management: Lucknow, Uttar Pradesh.

01-09-2007 to 23-07-2009 | **Assistant Professor**

(Electronics and Communication Engineering)

Shri Ramswaroop Memorial College of Engineering and Management: Lucknow, Uttar Pradesh, India

23-09-2002 to 31-08-2007 | **Lecturer**

(Electronics and Communication Engineering)

Professional Education

- 2019 **Ph.D.** Shri Ramswaroop Memorial University
(Broad-band EM Absorber based on Metamaterial)
Date of enrollment 11-07-2014, Date of completion 19-11-2019
<http://hdl.handle.net/10603/287131>
<https://shodhganga.inflibnet.ac.in/handle/10603/287131>
- 2008 **Master of Technology**, U.P.T.U. (AKTU) (Digital Communication, CPI-9.67)
- 2002 **Bachelor of Technology**, J.K. Institute of Applied Physics and Technology
Central University of Allahabad
(Electronics and Telecommunication, 72%)
- 1998 **Bachelor of Science** (Gold Medalist), Central University of Allahabad
(PCM Group, 77%)
- 1994 **All India Senior School Certificate Examination**, CBSE board
(PCM, Biology, English, 81%) - Batch Topper
- 1992 **All India Secondary School Certificate Examination**, CBSE board
(English, Hindi, Maths, Science, Social Science, 84%) – Batch Topper

Topic of Research (Ph.D)

‘Broad-band EM Absorber based on Metamaterial’

Guide: Dr. Mukul Misra (Director: Research and Consultancy)
Shri Ramswaroop Memorial University

Topic of M.Tech Dissertation

‘Performance Improvement of Long Haul Return-to-Zero Dense WDM Systems via Dispersion Interleaving’.

Guide:

Dr. Neelam Srivastava - HoD, Deptt. of ECE, Institute of Engineering and Technology (IET), Lucknow

Dr. Ragini Tripathi - HoD, Deptt. of EI, NIEC, Lucknow

Research Interests

Metamaterial based Electromagnetic Microwave Absorbers

Metamaterials (MM) have attracted researchers in recent years due to their potential applications in realization of novel devices such as perfect lens, cloak at microwave frequency, microwave antennas. Metamaterial concept is also applied for the development of the Metamaterial Perfect Absorbers (MPAs). With the demonstration of first MPA in 2008, significant research work has been carried out to improve the designing and performance of the MPAs maintaining the conventional three layered geometry. MPAs find applications in suppressing the electromagnetic interference, improving performance of radar system by reducing the RCS (radar cross section) and improving radiation pattern of transmitting antenna by suppressing side-lobes. Future applications include the use of THz MPA as frequency selective thermal detectors, spectrum modifiers, MMA based tunable micro-bolometer, Metamaterial THz waveguide. At present the research work focuses on MPAs with narrow band absorption, Polarization independent MPAs, MPAs with high absorption at wide angle of incidence of electromagnetic waves, MPAs with broad-band absorption.

In the research work three MPA structure designs were simulated, fabricated, and tested for maximum absorptance at resonance frequencies for normal incidence, for wide incidence angle and for different polarization angle of EM waves.

First MPA design: Oblique Incidence and Polarization Insensitive Multi-band Metamaterial Absorber with Quad Paired Concentric Continuous Ring Resonators.

Second MPA design: Wide Incidence Angle and Polarization Insensitive Dual Broad-Band Metamaterial Absorber based on Concentric Split and Continuous Rings Resonator Structure.

Third MPA design: Angle of Incidence and Polarization Insensitive Ultra-thin Ultra Broad-band Metamaterial Microwave Absorber based on Non-linearity in Unit Cell Design.

Teaching Experience (23+ years)

Subjects taught in U.G Programme

- **Digital Electronics**
- **Signals and Systems**
- **Digital Communication**
- **Analog Communication**
- **Optical Fibre Communication**
- **Computer Organization & Architecture**
- **Solid State Devices and Circuits**
- **Data Computer Networks**
- **Communication Engineering**

Subjects taught in P.G Programme

- **Information Theory and Coding**
- **Estimation and Detection Theory**

Subjects taught in P.hD. Course Work

- **Metamaterials and Metamaterial Perfect Absorber**
- **Scilab**
- **LabVIEW**

Software Skills

- | | |
|-------------------------------|-------------------------------|
| ➤ CST Microwave Studio | ➤ Optisystem |
| ➤ LabVIEW | ➤ Electronic Workbench |
| ➤ Matlab | ➤ Microsoft Visio |
| ➤ Scilab | ➤ HTML |
| ➤ C-Language | ➤ CSS |
| | ➤ JavaScript |

Lectures on my YouTube Channel

ETronicsLecturesbyDrAlkesh Agrawal

Subject: Digital System Design/Digital Electronics, Analog and Digital Communication, Scilab, LabVIEW, Metamaterial absorber design on CST Microwave studio.

Some YouTube Lecture Links:

https://youtu.be/5CU9NdKyAIA	(Logic Gates Universal Gates De-Morgan Laws)
https://youtu.be/YXYjJ7DjvI8	(NAND-NOR Implementation of Boolean Function)
https://youtu.be/CYKPAZZifxo	(Standard Canonical Forms SOP POS)
https://youtu.be/dQKDDplodMw	(5-Variable K-Maps)
https://youtu.be/ZXq-x9uh1Gc	(6-Variable K-Map)
https://youtu.be/QcycSRvbONU	(Combinational Circuit - Half Adder and Full Adder)

Developed Webpage based monthly Newsletter: *e-Lectronic Patrika*

<https://alkesh-agrawal.github.io/e-Newsletter-DECE/>

Research Publications in Refereed Journals (WoS and SCOPUS)

1. Saifullah Khalid, Sashi Kant Gupta, Midhun Chakkaravarthy, Dharmendra Prakash, **Alkesh Agrawal**, D. K. Nishad, "AI-Optimized UAV Swarm System for Radiation Fog Dispersal and Runway Visibility Enhancement at Airport", *International Journal of Aeronautical and Space Science*, 2025.
ISSN: 2093-2480 (I.F: 1.6) (WoS-SCIE)
<https://doi.org/10.1007/s42405-025-01097-0>
2. Sunil K. Dubey, Ashok Kumar Shankhwar, Nand Kishore, and **Alkesh Agrawal***, "High Gain and Bandwidth Enhanced Microstrip Patch Antenna with Defective Ground Structure Loaded with Metamaterial Unit Cells for Intelligent Transportation Systems," *Progress In Electromagnetics Research M*, Vol. 136, 57-67, 2025.
ISSN: 1937-8726 (IF: 0.8) (WoS-SCIE)
[doi:10.2528/PIERM25101001](https://doi.org/10.2528/PIERM25101001)
3. Bhagwant Singh, **Alkesh Agrawal***, Supriya Mishra, Alka Yadav, "Quad Circular with Plus Shaped Resonators Based Multi-Band Metamaterial Absorber", *International Journal of Microwave and Optical Technology*, Vol. 20, no. 3, pp. 243-250, Jun., 2025.
ISSN: 1553-0396 (Scopus Indexed)
4. Rahul Misra, Kunal Singh, **Alkesh Agarwal**, Ravi Rastogi, Sarvesh Dubey, "Electronic Noise Analysis of Source-Engineered Phosphorene/Si Heterojunction Dopingless Tunnel-FET," *Silicon* (Springer), July 16, 2022.
ISSN: 1876-9918 (IF: 2.67) (WoS-SCIE)
<https://doi.org/10.1007/s12633-022-02019-5>
5. **Alkesh Agrawal** and Mukul Misra, "Angle of Incidence and Polarization Insensitive Ultra-Thin Ultra Wide-Band Metamaterial Absorber based on Novel Non-Linearity in Unit Cell Design," *Journal of Microwaves, Optoelectronics and Electromagnetic Applications*, Vol. 20, No. 2, 556-569, 2021.
ISSN: 2179-1074 (Cite Score: 1.30) (Scopus Indexed)
<https://doi.org/10.1590/2179-10742021v20i31207>
6. Kunal Krishna Upadhyay, **Alkesh Agrawal***, and Mukul Misra, "Wide-Band Log-Periodic Microstrip Antenna with Defected Ground Structure for C-Band Applications," *Progress In Electromagnetics Research C*, Vol. 112, 127-137, 2021.
ISSN: 1937-8718 (Cite Score: 2.8) (Scopus Indexed)
<https://www.jpier.org/PIERC/pier.php?paper=21031106>
7. **Alkesh Agrawal**, Mukul Misra and Ashutosh Singh, "Wide incidence angle and polarization insensitive dual broad-band metamaterial absorber based on concentric split and continuous

- rings resonator structure”, *Material Research Express (IOP Science)*, vol. 5, no. 11, pp. 115801, 2018. ISSN: 2053-1591 (IF: 1.151) (WoS-SCIE)
<https://iopscience.iop.org/article/10.1088/2053-1591/aadd83>
8. **Alkesh Agrawal**, Mukul Misra, and Ashutosh Singh “Oblique Incidence and Polarization Insensitive Multiband Metamaterial Absorber with Quad Paired Concentric Continuous Ring Resonators,” *Progress In Electromagnetics Research M*, vol. 60, pp. 33–46, 2017.
 ISSN: 1937-8726 (IF: 2.949) (UGC Ref. no. 31874) (WoS-SCIE)
<http://www.jpier.org/PIERM/pier.php?paper=17061302>
 9. **Alkesh Agrawal**, Ashutosh Singh, and Mukul Misra, “A Multiband Metamaterial Absorber with Concentric Continuous Rings resonator Structure”, *International Journal of Advances in Microwave Technology (IJAMT)*, vol. 1, no.1, pp. 5-9, May 2016.
 ISSN: 2456-4346
 10. Varsha Pandey, **Alkesh Agrawal**, Mohd. Maroof Siddiqui, “Sleep Disorders and EEG Recording”, *International Journal of Electronics and Computer Science Engineering (IJECS)*, vol. 4, no. 3, pp. 207-212, 2015.
 ISSN: 2277-1956

Research Publications/Presentations in International Conf./IEEE Xplore (Indexed in SCOPUS)

1. Bhagwant Singh, Alkesh Agrawal, “Multi-band High Absorptance Split Outward “C” Shaped Resonators based Microwave Absorber for 5G Wi-Fi/ 5G Mid-band/ C-band / X-band Applications,” Anusandhan National Research Foundation (ANRF), Ministry of Science & Technology, Government of India Sponsored, International Conference on Innovations in Nanomaterials, Nano Devices & Smart Technologies (INNOST NANO 2025), 12-14 Nov., 2025.
<https://www.rwconferences.com/innost-nano-2025>
2. Alka Yadav, Alkesh Agrawal, “Polarization and Angle of Incidence Insensitive Triple-Band Metamaterial Absorber For S-, C- And X- Band Applications,” Anusandhan National Research Foundation (ANRF), Ministry of Science & Technology, Government of India Sponsored, International Conference on Innovations in Nanomaterials, Nano Devices & Smart Technologies (INNOST NANO 2025), 12-14 Nov., 2025.
<https://www.rwconferences.com/innost-nano-2025>

3. Bhagwant Singh, Alkesh Agrawal, “C and Inverted-C Shaped Multi-band Metamaterial Absorber”, 2024 International Conference on Control, Computing, Communication and Materials (ICCCCM), Prayagraj, U.P., India, Aug. 10-11, 2024. IEEE Conference Record No. 61016, Published in *IEEE Xplore*, pp. 467-470, Jun. 24, 2025.

DOI: [10.1109/ICCCCM61016.2024.11039930](https://doi.org/10.1109/ICCCCM61016.2024.11039930)

4. Khalid, S., Chakkaravarthy, M., Gupta, S. K., Prakash, D., **Agrawal, A.**, & Nishad, D. K. (2025). A Conceptual Framework for Developing an AI-Powered UAV System for Effective Fog Management at Airports. *SPAST Reports*, Vol. 2(1), LGPR Conference, 2025.

<https://spast.org/ojspace/article/view/5199>

5. Dharmendra Prakash, Alkesh Agarwal, Saifullah Khalid, Dinesh Kumar Nisha, Vinay Kumar Singh, "Innovative Approaches to Fog Dispersal: The Emerging Role of Unmanned Aerial Vehicles," 2024 IEEE 11th Uttar Pradesh Section International Conference on Electrical, Electronics and Computer Engineering (UPCON), Lucknow, India, Nov. 29 – Dec. 01, 2025. IEEE Conference Record No. 62832, Published in *IEEE Xplore*, pp. 1-6, May 12, 2025.

Electronic ISBN: 979-8-3503-7872-6

DOI: [10.1109/UPCON62832.2024.10983532](https://doi.org/10.1109/UPCON62832.2024.10983532).

6. Kunal Krishna Upadhyay, **Alkesh Agrawal** and Mukul Misra “Wide-band CPW-Fed Slotted Rectangular Patch Antenna loaded with Ring Resonators for C-band Applications” 2019 International Conference on Cutting-edge Technologies in Engineering (ICon-CuTE), Shri Ramswaroop Memorial University, Lucknow-Deva Road, Barabanki, November 16-18, 2019. IEEE Conference Record No. 47290. Published in *IEEE Xplore*, pp. 121 – 124, Feb. 13, 2020.

Electronic ISBN: 978-1-7281-0000-5

DOI: [10.1109/ICon-CuTE47290.2019.8991515](https://doi.org/10.1109/ICon-CuTE47290.2019.8991515)

7. **Alkesh Agrawal**, Ashutosh Singh, and Mukul Misra “A Dual Broadband Metamaterial Absorber with Concentric Continuous and Split Rings Resonator Structure” 3rd IEEE Uttar Pradesh Section International Conference on Electrical, Computer and Electronics Engineering (UPCON-2016), IIT- BHU, Varanasi, December 9-11, 2016. Published in *IEEE Xplore*, pp. 597 – 601, April 12, 2017.

Electronic ISBN: 978-1-5090-5384-1

DOI: [10.1109/UPCON.2016.7894722](https://doi.org/10.1109/UPCON.2016.7894722)

8. Dinesh Kumar, **Alkesh Agrawal**, Gopal Singh Phartiyal, “Non Iterative LDPC Codes Decoding by Syndrome Generation using Artificial Neural Network”, National conference on Recent Advances in Electronics and Computer Engineering (RAECE-15), IIT Roorkee, Feb. 13-15, 2015. Published in *IEEE Xplore*, pp. 22-25, July 14, 2016.

Electronic ISBN: 978-1-5090-2146-8

Book Chapter (Indexed in WoS and SCOPUS)

1. **Alkesh Agrawal**, "An Application-Based Study on Electromagnetic Absorber Using Metamaterial", in *Metamaterials - History, Current State, Applications, and Perspectives* [Working Title]. London, United Kingdom: IntechOpen, 2022 [Online].
ISBN: 978-1-80356-810-2
doi: 10.5772/intechopen.108146
<https://www.intechopen.com/online-first/84434>
2. **Alkesh Agrawal**, Bhagwant Singh, "Study of Novel Design of Multi-Band and Broad-Band Metamaterial Microwave Absorber", in *Metamaterial Technology and intelligent Metasurface for Wireless Communication Systems*. IGI Global, USA, 2023 [Online].
ISBN13: 9781668482872 | ISBN10: 1668482878 | EISBN13: 9781668482889
DOI: 10.4018/978-1-6684-8287-2.ch006
https://www.igi-global.com/book/_/312573
3. Dharmendra Prakash, **Alkesh Agrawal**, Saifullah Khalid, "Comparative Analysis of UAVs: A Study on Payload Capacity and Performance", in *New Innovations in AI, Aviation, and Air Traffic Technology*. IGI Global, USA, 2024 [Online].
ISBN(hc) 9798369319543 | ISBN(sc) 9798369350195 | eISBN 9798369319550
DOI: 10.4018/979-8-3693-1954-3.ch007
<https://www.igi-global.com/chapter/comparative-analysis-of-uavs/350534>
4. Dharmendra Prakash, **Alkesh Agrawal**, Saifullah Khalid, "A Comprehensive and In-Depth Review of Innovative Antenna Designs for UAVs Advanced Antenna Technologies for Aerial Platforms: From Design to Deployment". IGI Global, USA, 2025 [Online].

ISBN13: 9798369360354 | EISBN13: 9798369360378
DOI: 10.4018/979-8-3693-6035-4.ch001
<https://www.igi-global.com/chapter/a-comprehensive-and-in-depth-review-of-innovative-antenna-designs-for-uavs/375719>

5. S. Khalid, Shashi Kant Gupta, Midhun Chakkaravarthy, Dharmendra Prakash, **Alkesh Agrawal**, and Dinesh Kumar Nishad, "Graphene in Electromagnetic Interference Shielding: From Fundamentals to Aerospace and 5G Applications". IGI Global, USA, 2025 [Online].
ISBN13: 9798337338620 | EISBN13: 9798337338644
DOI: 10.4018/979-8-3373-3862-0.ch007
<https://doi.org/10.4018/979-8-3373-3862-0.ch007>
6. S. Khalid, Shashi Kant Gupta, Midhun Chakkaravarthy, Dharmendra Prakash, **Alkesh Agrawal**, and Dinesh Kumar Nishad, "Nano-Engineered Surface Coatings for Corrosion Resistance in UAV-Based Fog Dispersal Systems". IGI Global, USA, 2025 [Online].
ISBN13: 9798337338620 | EISBN13: 9798337338644
DOI: 10.4018/979-8-3373-3862-0.ch005
<https://doi.org/10.4018/979-8-3373-3862-0.ch005>

Patents Granted

1. Invention: AI-Driven UAV Seeding System for Enhanced Atmospheric Visibility
UK Patent Design No. 6422703

International Design Classification:

Version: 15-2025

Class: 12 Means of Transport or Hoisting

Subclass: 07 Aircraft and Space Vehicles

2. Invention: Hybrid UAV with AI-Based Flight optimization of Fog clearance

UK Patent Design No. 6422704

International Design Classification:

Version: 15-2025

Class: 12 Means of Transport or Hoisting

Subclass: 07 Aircraft and Space Vehicles

3. Invention: UV Ray Generator-Integrator UAV for Fog Dispersal

UK Patent Design No. 6422705

International Design Classification:

Version: 15-2025

Class: 12 Means of Transport or Hoisting

Subclass: 07 Aircraft and Space Vehicles

4. Invention: Portable UAV-Mounted UV Ray Generator for Atmospheric Modifications

UK Patent Design No. 6422706

International Design Classification:

Version: 15-2025

Class: 12 Means of Transport or Hoisting

Subclass: 07 Aircraft and Space Vehicles

5. Invention: High Precision UV Ray Emission Apparatus for Drones

UK Patent Design No. 6422707

International Design Classification:

Version: 15-2025

Class: 12 Means of Transport or Hoisting

Subclass: 07 Aircraft and Space Vehicles

6. Invention: AI-Controlled UV Ray Distribution Apparatus for UAVs

UK Patent Design No. 6422708

International Design Classification:

Version: 15-2025

Class: 12 Means of Transport or Hoisting

Subclass: 07 Aircraft and Space Vehicles

7. Invention: Autonomous Fog Identification and Tracking Apparatus for UAVs

UK Patent Design No. 6422709

International Design Classification:

Version: 15-2025

Class: 12 Means of Transport or Hoisting

Subclass: 07 Aircraft and Space Vehicles

8. Invention: Multi-Spectral Imaging Device for Real-Time Fog Detection and Mapping

UK Patent Design No. 6422710

International Design Classification:

Version: 15-2025

Class: 14 Recording, Telecommunication or Data Processing Equipment

Subclass: 01 Equipment for the Recording or Reproduction of Sounds or Pictures

9. Invention: Multi-Purpose UAV for Meteorological Data Collection & Dispersal Operations

UK Patent Design No. 6422711

International Design Classification:

Version: 15-2025

Class: 12 Means of Transport or Hoisting

Subclass: 07 Aircraft and Space Vehicles

10. Invention: AI-Enabled UAV for Automated Fog Dispersal & Visibility Enhancement

UK Patent Design No. 6422712

International Design Classification:

Version: 15-2025

Class: 12 Means of Transport or Hoisting

Subclass: 07 Aircraft and Space Vehicles

Patents Published/Filed

1. **Invention:** IoT based multi-tasking ROBOT. (Filed)

Patent Application No. 202511023726 A

Date of Filing: 17/03/2025

2. **Invention:** Microstrip Patch Antenna with Metamaterial Unit Cells Loaded Radiating Patches and Ground Structure for Its Applications. (Published)

Patent Application No. 202511024894 A

The Patent Office Journal No. 14/2025 Dated 04/04/2025

3. **Invention:** Robotic Wheel Chair for Physically disabled. (Published)

Patent Application No. 202311011812 A

The Patent Office Journal No. 34/2024 Dated 23/08/2024

4. **Invention:** Automated engine seize system using over consumed alcohol sensors to prevent road accidents. **(Published)**
Patent Application No. 202211058007 A
The Patent Office Journal No. 15/2024 Dated 12/04/2024
5. **Invention:** Dual power mode sensor based automated low cost SMART Bin under Swachh Bharat Abhiyan. **(Published)**
Patent Application No. 202211052875 A
The Patent Office Journal No. 12/2024 Dated 22/03/2024
6. **Invention:** Ultra-Thin Ultra Broad-band Metamaterial Microwave Absorber Based on Novel Non-Linearity in Unit Cell Design for X-band Applications. **(Published)**
The Patent Office Journal No. 37/2021 Dated 10/09/2021
Patent Application No. 202011009840 A
7. **Invention:** De-Centralized Solar Charge cum Docking Station for 48Vdc Electric Vehicles. **(Published)**
The Patent Office Journal No. 25/2020 Dated 19/06/2020
Patent Application No. 201911044345 A

Paper Presentations in National and International Conferences

1. Alka Yadav, **Alkesh Agrawal**, “Triple-band High Absorptance Microwave Aborber for Solar Applications”, International Conference on Sustainable Innovation: Bridging Research & Industry for a Resilient Future, pp. 99, 2025. ISBN: 978-93-342-6452-4
2. Saumya Shukla, **Alkesh Agrawal**, “Designing Renewable-Powered Antennas for Energy Saving Wireless Networks”, International Conference on Sustainable Innovation: Bridging Research & Industry for a Resilient Future, pp. 100, 2025. ISBN: 978-93-342-6452-4
3. Dharmendra Prakash, **Alkesh Agrawal**, Saifullah Khalid, “Innovative Approaches to Fog Dispersal: The Emerging Role of Unmanned Aerial Vehicles”, 11th IEEE UP Section International Conference on Electrical, Electronics and Computer Engineering (UPCON-2024), SRMCEM, India, Nov. 29- Dec.01, 2024
4. **Alkesh Agrawal** and Bhagwant Singh “C and Inverted C Shaped Multi-band Metamaterial Absorber”, 3rd International Conference on Control, Computing, Communication and Materials, United College of Engineering & Research, Prayagraj, India, August 10-11, 2024.
5. Kunal Krishna Upadhyay, **Alkesh Agrawal** and Mukul Misra “Wide-band CPW-Fed Slotted Rectangular Patch Antenna loaded with Ring Resonators for C-band Applications” 2019 International Conference on Cutting-edge Technologies in Engineering (ICon-CuTE), Shri Ramswaroop Memorial University, Lucknow-Deva Road, Barabanki, November 16-18, 2019. IEEE Conference Record No. 47290. Published in *IEEE Xplore*, pp. 121 – 124, Feb. 13, 2020.

6. **Alkesh Agrawal**, Ashish Anand and Mukul Misra, “LabVIEW Implementation of Hamming (n,k) Channel Coding Scheme”, Second International Conference on Computing, Communication, and Control Technology (IC⁴T 2018), organized by Shri Ramswaroop Memorial Group of Professional Colleges and Shri Ramswaroop Memorial University, Lucknow, Oct. 25-27, 2018.
7. **Alkesh Agrawal** and Mukul Misra, “Metamaterial Perfect Absorbers”, Second International Conference on Computing, Communication, and Control Technology (IC⁴T 2018), organized by Shri Ramswaroop Memorial Group of Professional Colleges and Shri Ramswaroop Memorial University, Lucknow, Oct. 25-27, 2018.
8. Divya Pandey, **Alkesh Agrawal** and Manju Bhardwaj, “Resource Optimization in C-RAN using D2D for 5G Wireless Communications”, Second International Conference on Computing, Communication, and Control Technology (IC⁴T 2018), organized by Shri Ramswaroop Memorial Group of Professional Colleges and Shri Ramswaroop Memorial University, Lucknow, Oct. 25-27, 2018.
9. **Alkesh Agrawal**, Ashutosh Singh, and Mukul Misra “A Dual Broadband Metamaterial Absorber with Concentric Continuous and Split Rings Resonator Structure” 3rd IEEE Uttar Pradesh Section International Conference on Electrical, Computer and Electronics Engineering (UPCON-2016), IIT- BHU, Varanasi, December 9-11, 2016. Published in *IEEE Xplore*, pp. 597 – 601, April 12, 2017.
10. Dinesh Kumar, **Alkesh Agrawal**, Gopal Singh Phartiyal, “Non Iterative LDPC Codes Decoding by Syndrome Generation using Artificial Neural Network”, National conference on Recent Advances in Electronics and Computer Engineering (RAECE-15), IIT Roorkee, Feb. 13-15, 2015. Published in *IEEE Xplore*, pp. 22-25, July 14, 2016.
11. **Alkesh Agrawal**, Ashutosh Singh, and Mukul Misra “Recent Trends and Applications of Metamaterial based Microwave Absorbers” International Conference on Computing, Communication and Control Technology (IC⁴T-2016), organized by Shri Ramswaroop Memorial Group of Professional Colleges and Shri Ramswaroop Memorial University, Lucknow, November 10-12, 2016.
12. F. Fatima, **Alkesh Agrawal**, T. B. Joseph, “Design and Fabrication of Multi-band Microstrip Patch Antenna”, National conference on Recent Advances in Electronics and Computer Engineering (RAECE-15), IIT Roorkee, Feb. 13-15, 2015.
13. **Alkesh Agrawal**, Ashutosh Singh, Mukul Misra, “A Multiband Metamaterial Absorber using Concentric Circular Ring Shaped Structure”, 10th International conference on Microwave Antenna, Propagation and Remote Sensing (ICMARS-14), organized by International Centre for Radio Waves, Jodhpur, India, Dec. 09-12, pp-328-331, 2014.
14. Geetanjali Srivastava, **Alkesh Agrawal**, “Performance comparison of Gold codes and Walsh codes in DS-SS System,” in National Seminar on **Communications 2020** conducted by IETE & IE (India) held on Mar. 8-9, 2008.

15. **Alkesh Agrawal**, Neelam Srivastava, Ragini Tripathi, “Performance Improvement of Long Haul Return-to-Zero Dense WDM Systems via Dispersion Interleaving,” in National Seminar on **Communication Convergence** conducted by IETE & IE (India) held on Sep. 8-9, 2007.

FDPs/Workshops/VACs Organized

1. **Resource Person:** Guest Lecture on “How to file Patent Online – A case Study”, on the occasion of World Intellectual Property Rights (IPR) Day, organized by Research and Consultancy, SRMU, April 26th, 2025.
2. **Convener:** National Workshop on “**Role of AI in Achieving the Sustainable Development Goals (SDGs)**”, organized from Nov. 5 – Nov. 16, 2024 by Department of Electrical and Electronics Engineering, Shri Ramswaroop Memorial University, Barabanki, 225003.
3. **Resource Person:** 01 week FDP on “Hands-on LabVIEW Programming” organized from Jan. 09 – Jan. 13, 2024 by Department of Electrical and Electronics Engineering, Shri Ramswaroop Memorial University, Barabanki, 225003.
4. **Resource Person:** 01 week FDP for faculty members of Electronics and Communication Engineering and Electrical Engineering on “Hands-on Training on NI-LabVIEW” organized from Aug. 05 – Aug. 12, 2023 by Department of Electronics and Communication Engineering, Shri Ramswaroop Memorial Group of Professional Colleges, Tewariganj, Lucknow.
5. **Convener and Resource Person:** 04 weeks Summer Internship on “Hands-on Training on NI-LabVIEW and MyRIO” organized from June 21 – July 20, 2023 by Department of Electronics and Communication Engineering, Shri Ramswaroop Memorial University, Barabanki, 225003.
6. **Convener and Resource Person:** 01 week FDP on “Hands-on LabVIEW Programming” organized from Jul. 12 – Jul. 16, 2023 by Department of Electronics and Communication Engineering, Shri Ramswaroop Memorial University, Barabanki, 225003.
7. **Convener and Resource Person:** 01 day Workshop on “Implementation of Channel Coding Schemes in LabVIEW & Scilab” organized on April 29, 2023 by Department of Electronics and Communication Engineering, Shri Ramswaroop Memorial University, Barabanki, 225003.
8. **Convener and Resource Person:** 05 days FDP on “Basics of LabVIEW” organized from Jan. 25 – Jan. 30, 2023 by Department of Electronics and Communication Engineering, Shri Ramswaroop Memorial University, Barabanki, 225003.
9. **Convener and Resource Person:** 30 hours value added course on “Basics of HTML and CSS with application to build an effective personalized CV and web page layout (Basic Module)”, Aug. 01 – Dec. 08, 2022.

10. **Convener and Resource Person:** 06 days FDP on “Basics of Scilab” organized from June 21 – June 25, 2022 by Department of Electronics and Communication Engineering, Shri Ramswaroop Memorial University, Barabanki, 225003.
11. **Convener and Resource Person:** 06 days FDP on “Applications of Scilab” organized from July 05 – July 09, 2022 by Department of Electronics and Communication Engineering, Shri Ramswaroop Memorial University, Barabanki, 225003.
12. **Convener:** 06 days short term training course/workshop on “Verilog HDL using Xilinx ISE 14.5”, organized in Shri Ramswaroop Memorial University, Lucknow-Deva Road, Oct. 09 – 14, 2018.
13. **Convener:** 36 hours value added course on LabVIEW Programming (Basic Module), Sep. 13 – Dec. 30, 2018.
14. **Co-convenor:** 06 days short term training course on “Designing, Simulation, and Modeling of Electronic Circuits”, Sep. 18 – 23, 2017.
15. **Co-convenor and Resource Person:** 06 days short term training course on “MATLAB Programming: From Introduction to Essentials”, March 25 – 31, 2017.

Responsibilities in IEEE Conferences

1. **Organizing Chair** in 2019 International Conference on Cutting-edge Technologies in Engineering organized by Shri Ramswaroop Memorial University, Nov. 16 – Nov. 18, 2019. (ICon-CuTE-2019)
Conference Record No. 47290
<http://icon-cute-2019.srmu.ac.in/>
2. **Session Chair** in 11th IEEE UP Section International IEEE Conference on Electrical, Electronics and Computer Engineering (UPCON-2024), 29th Nov. – 01st Dec., 2024.
Conference Record No. 62832

Offline Trainings/FDPs /Workshops/Seminars Attended

1. One week Workshop on “Latest Trends in Research and Innovation in Higher Education” organized by Research Cell SRMU, Jan. 08 – Jan. 12, 2024.
2. One week Workshop on “Application of Intellectual Property Rights in Science & Technology” organized by IPR Cell SRMU, Feb. 20 – Feb. 24, 2023.
3. 02 days FDP on **Resource Mobilization for Research**, organized by IQAC, Shri Ramswaroop Memorial University, July 15 – July 16, 2019.
4. 06 days FDP on **Python with Data Science**, organized by IQAC, Shri Ramswaroop Memorial University, July 16 - July 20, 2019.
5. 02 days National Seminar on **Use of Technical Terminology in Science and Technology**, organized by Commission for Scientific and Technical Terminology,

ministry of Human Resource Development, Department of Higher Education, Government of India, Oct. 23- Oct. 24, 2019.

6. One week FDP on **“Challenges of Assuring Quality in Higher Education and Research”**, from July 02 – July 07, 2018, organized by IQAC, Shri Ramswaroop Memorial University, Lucknow-Deva Road, Barabanki.
7. One day **IP Awareness Programme on National IPR Policy 2016**, organized by Intellectual Property Office, India at Shri Ramswaroop Memorial University on Aug. 08, 2018.
8. 06 days FDP on **New Modes of Learning and Teaching in Higher Education**, organized by IQAC, Shri Ramswaroop Memorial University, July 10 - July 15, 2017.
9. 06 days FDP on **Quality Enrichment in Higher Education: Strategies towards Continual Improvements in Education System and R & D**, organized by IQAC, Shri Ramswaroop Memorial University, July 11- July 16, 2016.
10. 06 days FDP on **Effective Teaching-Learning Process**, conducted by Shri Ramswaroop Memorial University, July 6- July 11, 2015.
11. **Workshop on Electromanetics and Antenna Design (WEAD-2016)** using ANSYS HFSS conducted by IIT BHU on 9th and 10th of April 2016.
12. 06 days FDP on **Challenges and Issues in Higher Education**, conducted by Shri Ramswaroop Memorial University, June 30- July 04, 2014.
13. **Dale Carnegie** training workshop on **“High Impact Teaching Skills”** conducted by Wipro on 26th and 30th of October, 2009.
14. Workshop on **‘VLSI Design Tools’** conducted by Deptt. of Electronics Engineering IET Lucknow on 17-18 Nov. 2007.
15. Seminar on **‘Shaping Young Minds’** conducted by Lucknow Management Association, July 2007.

Online Trainings/FDPs /Workshops/Webinars Attended

1. Innovation Ambassador (IA) training ‘Foundation Level’ (15 sessions) conducted by MoE’s Innovation Cell & AICTE. Certificate No. IA/Foundation/1013134, 15-10-2022. IIC ID: IC202115790.
2. 6th Annual Library Technology Conclave 2022 On Research Data Management and Stewardship: Building Blocks of Open Science, April 28th – 30th 2022, organized by Somaiya Vidyavihar University, Mumbai, Informatics Limited, Bangalore, MYRA School of Business, Mysore.
3. One week online FDP on **“Issues, reforms and related guidelines for Higher Education Institutes”**, from Aug. 16 – Aug. 21, 2021, organized by IQAC, Shri Ramswaroop Memorial University, Lucknow - Deva Road, Barabanki.

4. One week online FDP on “**ICT Enabled Tools for Effective Teaching-Learning, Examination Processes in Higher Education**”, from June 16 – June 20, 2020, organized by IQAC, Shri Ramswaroop Memorial University, Lucknow-Deva Road, Barabanki.
5. Online one week FDP from May 11 – May 16, 2020 on ‘**R**’ organized by Birla Institute of Technology Mesra, Noida Campus in association with Spoken Tutorial Project, IIT Bombay, funded by the National Mission on Education through ICT, MHRD, Govt. of India.
6. Online one week FDP from May 21 – May 26, 2020 on ‘**Scilab**’ organized by Pimpri Chinchwad College of Engineering, Nigdi, Pune in association with Spoken Tutorial Project, IIT Bombay, funded by the National Mission on Education through ICT, MHRD, Govt. of India.
7. Online one week FDP from June 01 – June 06, 2020 on ‘**Scilab**’ organized by Integrated Intelligent Research (IIR), Chennai in association with Spoken Tutorial Project, IIT Bombay, funded by the National Mission on Education through ICT, MHRD, Govt. of India.
8. 02 Days online National Workshop on “Robotics and Control,” organized by Department of Electronics Engineering, Madhav Institute of Technology and Science, Gwalior on 16-17 May, 2020.
9. Online Webinar on “Relevance of IEEE Standards in Teaching, Learning and Industry Collaborations,” organized by EBSCO and IEEE India Services.
10. Online Webinar on “How To Publish a High Quality Technical Journal Paper,” organized by IEEE India Services.
11. Online Webinar on “Leveraging Technology in Education,” organized by TMH.
12. Online Webinar on “Python World,” organized by TMH.
13. Online Webinar on “Online Teaching-Learning & Assessment,” and “NAAC Accreditation Management System,” organized by Inpods.
14. One week online winter FDP on **Wireless and Mobile Communication** from 03-12-18 to 07-12-18 (in association with principal coordinating academy-IIT Guwahati and co-principal coordinating academy-NIT Patna, under Ministry of Electronics and Information Technology (MeitY)).
15. One week online winter FDP on **DSP and Sensors** from 10-12-18 to 14-12-18 (in association with principal coordinating academy-NIT Warangal and co-principal coordinating academy-MNIT Jaipur, under Ministry of Electronics and Information Technology (MeitY)).
16. One week online winter FDP on **AI and Machine Learning** from 17-12-18 to 21-12-18 (in association with principal coordinating academy-IIT Roorkee and co-principal coordinating academy-IIITDM Jabalpur, under Ministry of Electronics and Information Technology (MeitY)).

Certification Courses

1. **Udemy Online Course**- Build Responsive Real-World Websites with HTML and CSS, (37.5 Hours), May 27, 2022.
2. **Udemy Online Course**- The Complete 2022 Web Development Bootcamp, (55.5 Hours), March 20, 2022.
3. **Publons (WoS) Academy** Practical Peer Review Certification Course, June 01, 2021.
4. **JCR-South Asia Training and Certification** Program 2020, organized by WoS: A Clarivate Analytics Company.
5. **E-Shodh Sindhu WoS** Certification Series, organized by INFLIBNET Center and Clarivate Analytics.
6. On-line self paced training course on **MATLAB Onramp**, organized by Mathworks Services.
7. E-learning Certification Program on Industry 4.0, organized by TATA Steel, Capability Development.
8. E-learning Certification Program on Machine Learning, organized by TATA Steel, Capability Development.
9. E-learning Certification Program on Advanced MS-Excel, organized by TATA Steel, Capability Development.
10. On-line Courses on E-Mail Writing, Resume Writing, Interview Skills, Presentation Skills, organized by TCSiON.

M.Tech./M.Sc. Dissertation Supervised

1. Image Processing Applications using LabVIEW
(Mansi Pandey- 202310201010004 & Nashra Juned- 202310201010003)
2. Metamaterial Absorber for 5G Applications.
(Mr. Vikas Soni-202210201010001)
3. Study and Design of Triple-Band Microstrip Patch Antenna.
(Mr. Shashwat Srivastava -202110201010002)
4. Study and Design of Microstrip Patch Antenna with Directional Radiation Pattern.
(Mr. Virendra Kr. Yadav-202110201010003)
5. Study and Design of Multi-band Microwave Patch antenna.
(Mr. Shubham Chaudhry-201810104020801)
6. Study and Design of Wide-band Patch Antenna with DGS in C and X-band.
(Mr. Kunal Upadhyaya-201610104010004)
7. Study and Design of Metamaterial Absorber for Application in X-band.
(Ms. Farheen Fatima-201410104010004)

8. Study and Design of Metamaterial Absorber for Application in C-band.
(Ms. Priyanka Pandey-201410104010001)
9. Modification of 406 MHz Emergency Locator Transmitter for Lower False Alarm Rate and better Probability of Activation.
(Mr. Manish Kumar-201410104020801)
10. Noise Cancellation in Fetal Electrocardiography using Particle Swarm Optimization.
(Ms. Komal Dixit-201410104010002)
11. Image Domain Steganography using LSB Replacement Method.
(Mr. Bhagwant Singh-201310104010001)
12. Performance Analysis of Genetic Algorithm based LAS-CDMA Generation.
(Mr. Mukesh Pal-201310104010002)
13. Performance Analysis of IDMA Scheme using Quasi Cyclic Low density Parity Check Codes.
(Mr. Anurag saxena-201310104010004)
14. Non-iterative LDPC Codes Decoding by Syndrome generation using Artificial Neural Network.
(Mr. Dinesh Kumar-201210104010001)

Awards and Honor

1. **E-Learning Excellence Award**-2021 for use of ICT practices at Shri Ramswaroop Memorial University for academic session 2020-21 on June 18, 2022.
2. **Second best paper award** for paper presentation :
Alkesh Agrawal, Ashutosh Singh, and Mukul Misra “A Dual Broadband Metamaterial Absorber with Concentric Continuous and Split Rings Resonator Structure,” presented in 3rd IEEE Uttar Pradesh Section International Conference on Electrical, Computer and Electronics organized by IIT(BHU), Varanasi, December 9-11, 2016.
3. **Second best paper award** for paper presentation:
Alkesh Agrawal, Ashutosh Singh, and Mukul Misra “A Multiband Metamaterial Absorber using Concentric Circular Ring Shaped Structure”, presented in 10th International conference on Microwave Antenna, Propagation and Remote Sensing (ICMARS-14), organized by International Centre for Radio Waves, Jodhpur, India, December 09-12, 2014.
4. **Best Faculty Award** for academic year 2009-10 by SRM College of Engineering and Management.
5. **Appreciation Letter** for academic year 2008-09 by SRM College of Engineering and Management.
6. **Best Faculty Award** for academic year 2002-03 by SRM College of Engineering and Management.

7. **Suhrota Roy Memorial Scholarship** for securing highest marks in Physics and Maths in B.Sc.-I in yr. 1996.
8. **Malvea Memorial Scholarship** for securing highest marks in B.Sc.-I (PCM) in yr. 1996.
9. **Smt. Chandramani Agrawal Memorial Scholarship** for securing highest marks in Physics in B.Sc.-I & II, session 1997-98.
10. **Mitra Memorial Scholarship** for securing highest marks in Maths in B.Sc.-I & II, session 1997-98
11. **Ewing Christian College Gold Medal** for highest marks in B.Sc. (PCM/PSM/PEM group) 1995-98

Professional Memberships

1. **IEEE** Professional Membership – 94966964
2. Member **IAENG**
3. **E-Elsevier** Editorial Board Membership - EDWIN-003736603947

Other Responsibilities

Shri Ramswaroop Memorial University (2012 – till date)

- Appointed as Research Executive (IPR and Patents): 2025-28, Research and Consultancy, SRMU.
- IIC-SRMU, President.
- Officiating Director, Centre of Innovation, Incubation and Entrepreneurship, SRMU.
- Officiating Dean, Faculty of Electronics & Communication Engineering, SRMU.
- Faculty Co-ordinator, Faculty of Electronics & Communication Engineering, SRMU.
- Ex-officio Member, Board of Studies, FoECE, SRMU.
- Member, steering committee of IQAC, Criterion VII (Innovation and Best Practices).
- Member, Center of Energy Studies and Research, SRMU.
- Coordinator, Internal audit team, SRMU.
- Head Examiner, End Semester Examination (ESE), SRMU.
- Project coordinator
- Dissertation coordinator
- Head Observer, University Time Table, SRMU.

- Paper setter for End Semester Examination (ESE), SRMU.
- Time Table committee coordinator
- Coordinator Photography cell
- Coordinator Technical cell

Shri Ramswaroop Memorial College of Engineering & Management (2002 - 2012)

- ❖ Officiating HOD of EI Department.
- ❖ Deputed as **Center Controller for UP-SEE** in 2008 and 2009.
- ❖ **Center Coordinator** of M.Tech programme (UPTU) in the year 2008, 2009 and 2011.
- ❖ **Verifying Officer** for UPSEE-2011 Councelling.
- ❖ Time Table coordinator.

References

Dr. Arun Prakash

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